P6 Section 7 — Tic-Tac-Toe with Facial Controller

Emotions map to board indices: neutral→0, happy→1, surprise→2 (row then column). Run with `python run.py`.

How well did the interface work?

The webcam path crops a centered square, converts to grayscale, resizes to 150×150, expands to 3 channels, and runs `results/best_basic_model.keras`. The `text` override recovers from wrong detections.

Webcam accuracy vs test-set accuracy?

Held-out facial recognition test accuracy is **0.7678**. Webcam accuracy is usually a bit lower because of lighting/pose/framing domain shift.

If not, why not?

Training faces are tightly cropped and centered; webcam frames differ in illumination, expression intensity, and alignment.

_get_emotion implementation

```
def _get_emotion(self, img) -> int:
    img = cv2.resize(img, image_size)
    img = np.stack((img, img, img), axis=-1)
    img = np.expand_dims(img, axis=0)
    prediction = self.model.predict(img, verbose=0)
    pred = int(np.argmax(prediction[0]))
    # Keras dir order is alphabetical (happy,neutral,surprise)
    # Assignment order is neutral=0, happy=1, surprise=2
    mapping = {0: 1, 1: 0, 2: 2}
    return mapping[pred]
```

Example move trace

```
Move 1: X happy(1)+neutral(0) → (1,0)
Move 2: O random → (0,1)
Move 3: X surprise(2)+happy(1) → (2,1)
Move 4: O random → (0,0)
Move 5: X happy(1)+happy(1) → (1,1)
Move 6: O random → (2,0)
Move 7: X happy(1)+surprise(2) → (1,2) → X wins

Replace this with your live webcam trace before Canvas submit.
```